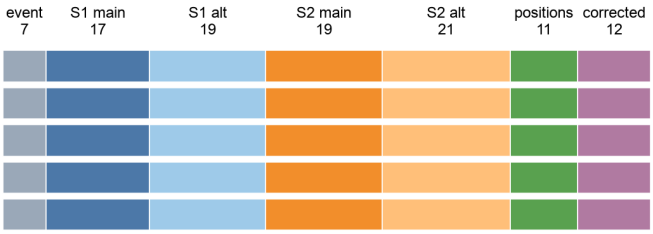
Proposal: a structured event data model in amstrax

For discussion with the XAMS group, 31-10-2026. No code yet. Status of amstrax master 761f62f; numbers from PROD3 runs 7790 (Cs, 97 V/cm) and 7912 (Cs, 50 V/cm).

In short. Replace the flat main / alternative columns by **two linked tables**: a slim **events** table (one row per event, ~28 columns instead of 104) and an **event_peaks** table (one row per peak inside an event, ~22 columns, any number of S1s and S2s). A loader gives the familiar flat table or a nested view (event → its S1s, S2s). Storage stays about the same; multiple scatters become straightforward.

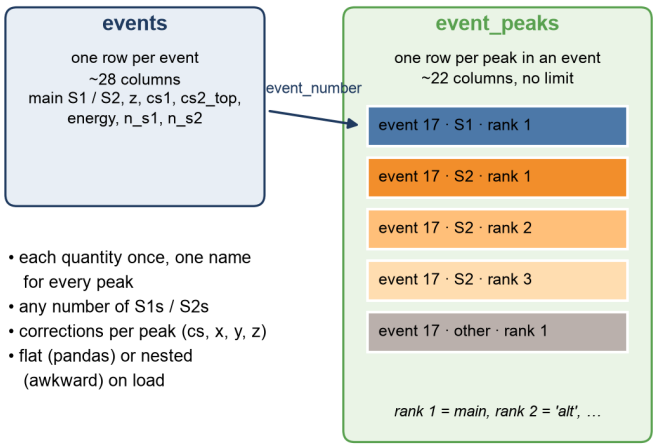
1 Now and proposed

now: event_info, one row per event, 104 columns



- the same 23 peak quantities 4× (main / alt S1, main / alt S2): 76 of the 83 event_basics columns
- room for 2 S1s and 2 S2s only: a 3rd S2 is lost
- corrections only for main and alt; names differ per peak (s2_area, alt_s2_area, …)

proposed: two linked tables



- each quantity once, one name for every peak
- any number of S1s / S2s
- corrections per peak (cs, x, y, z)
- flat (pandas) or nested (awkward) on load

Figure 1. Left: the 104 columns of event_info in blocks (proportional widths). Right: the proposed tables, linked by event_number; within an event the peaks of one type are ranked by area.

2 The two tables

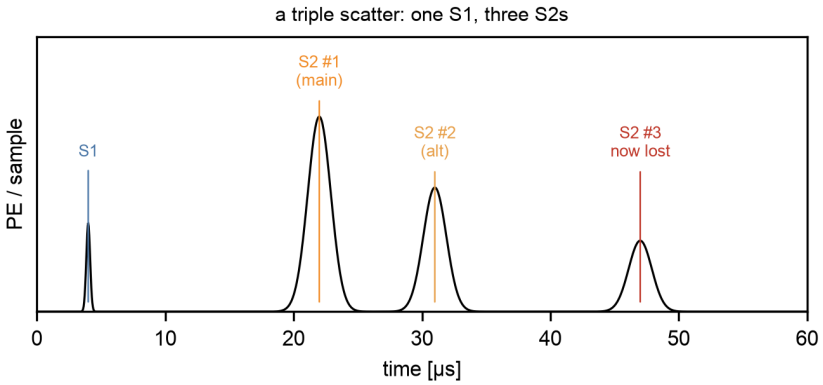
events (one row per event)	columns	vs now
identity	time, endtime, event_number	as now
content	n_peaks, n_s1, n_s2	n_s1 / n_s2 new
main S1	s1_index, s1_area, s1_area_fraction_top, s1_range_50p_area, s1_n_channels, s1_tight_coincidence	6 of 17
main S2	s2_index, s2_area, s2_area_fraction_top, s2_range_50p_area, s2_n_channels	5 of 19
interaction	drift_time, x, y, r, z	one x / y (was 3 copies)
corrected	cs1, cs2, cs2_top, energy	corrected_areas 1.0
quality	area_before_main_s2, large_s2_before_main_s2	as now

≈ 28 columns. Kept on purpose: everything a typical plot of the main interaction needs, so most analyses never touch the peak table.

event_peaks (one row per peak in an event)	columns
link	event_number, peak_index (row in peaks), type (1 S1, 2 S2, 0 other), rank (by area within type; 1 = main)
time	time, endtime, center_time, dt_s1 (time after the main S1: the drift time of an S2)
size / shape	area, area_fraction_top, range_50p_area, range_90p_area, rise_time
channels	n_channels, n_hits, max_pmt, max_pmt_area, tight_coincidence, n_saturated_channels
position (S2)	x, y, z (z of an S2 from dt_s1)
corrected	cs (S1: z and (x, y) of the main S2; S2: top array, own z and (x, y))

≈ 22 columns. 'alt' is no longer a separate set of names: the alternative S2 is the S2 with rank 2.

3 Example: a triple scatter



	now (event_basics)	proposed (event_peaks)
S1	s1_area …	row: S1, rank 1
S2 #1	s2_area …	row: S2, rank 1
S2 #2	alt_s2_area …	row: S2, rank 2
S2 #3	—	row: S2, rank 3

Figure 2. With two S2 slots the third interaction disappears from the event data; in event_peaks it is one more row.

4 How do I …

question	now (event_info)	proposed (ev = events, p = event_peaks)
main S2 area	`e['s2_area']`	`ev['s2_area'] (unchanged)`
2nd largest S2	`e['alt_s2_area']`	`p[(p.type == 2) & (p.rank == 2)]`
3rd largest S2	`not available`	`p[(p.type == 2) & (p.rank == 3)]`
all S2s > 500 PE	`not available`	`p[(p.type == 2) & (p.area > 500)]`
events with exactly two S2s	`approximate (alt_s2 > cut)`	`ev[ev.n_s2 == 2]`
nested, per event	–	`ev = load_events(run, nested=True); ev.s2.area[:, :3]`

5 Size and speed

	now	proposed
rows	1 per event	events: 1 per event; event_peaks: 3.3–3.5 per event (7790, 7912)
bytes per event	event_basics 360 + positions + corrected ≈ 470	events ≈ 120 + peaks 3.4 × ≈ 70 ≈ 360
loading the main interaction	one table	one (slim) table: faster
multiple-peak questions	not possible beyond 2	one filter / group-by on event_peaks

Both are ordinary strax data types (fixed columns, stored per chunk), so processing, storage and the dashboard machinery stay as they are; the nested view (awkward-array) is built only in the loader.

6 What changes for users, and migration

step	what
1	Agree on the column lists (this note) — which columns do the students actually use?
2	amstrax: plugins events (slim) and event_peaks; corrections per peak (same correction files as corrected_areas 1.0); loader load_events(run, nested=False / True).
3	Transition: event_info stays for one version next to the new tables (deprecated), so notebooks keep working.
4	Regenerate the event level of the versions in use (from stored peaks / event_basics, no raw data): snapshot first, then the shared checkout — no gap for users.
5	Dashboard (event view, dataview columns), handbook page, example notebook; then retire event_info.

Breaks: code that uses alt_* names or the 3 copies of x / y (the dashboard dataview, the event displays of the SR analyses). Does not break: peaks, peak_basics, records and everything below the event level.

7 Open questions for the discussion

question	options
rank by area, or by time?	area (as main / alt now); time order is the dt_s1 column
which peaks belong to an event?	all peaks inside the event window (as now), or only S1s before / S2s after the main S1
S1 correction for several S2s	with the position of the main S2 (as now), or one cs1 per S1-S2 pair
nested view	awkward-array (extra dependency) or pandas with a MultiIndex (event, peak)
keep event_info?	one version as a transition, or permanently as a convenience merge